

**Q. 1** (Ice cream). Thomas Sweet sells three flavors of ice cream: vanilla (\$1 per cone), chocolate (\$2 per cone), and strawberry (\$4 per cone). However, by the time you arrive at the store some of the flavors may have sold out. Each flavor is sold out independently with probability  $\frac{1}{2}$ .

- a. What is the probability that all flavors are sold out?
- b. If the store still has some ice cream when you get there, you buy one ice cream cone of the cheapest flavor that is available. What is the expectation of the cost of your cone, given that not all the flavors are sold out?
- c. You discover you are allergic to strawberries. Therefore, when you go to the store you buy ice cream only if either vanilla or chocolate flavors are available. Given that you do not buy an ice cream cone, what is the probability that all ice cream in the store was sold out?

**Q. 2** (Organic chemistry). Organic chemistry is a really hard subject, but it is so cool that 100 students sign up to take it. However, over the course of the semester, each student gets disgruntled independently with 1% probability. Disgruntled students sign a petition asking for the professor to be spoken to firmly by the university.

- a. What is the probability that exactly  $\frac{1}{4}$  of the class signs the petition?
- b. The university is so disturbed by the petition that they consider whether to fire the professor. The probability that the university decides to fire the professor is precisely the *square* of the fraction of students that sign the petition (for example, if half the students sign the petition, then the university decides to fire the professor with probability  $\frac{1}{4}$ ). What is the probability that the professor is fired?

**Q. 3** (Cupcakes). To celebrate the end of midterm week, you get two cupcakes in the dining hall today. However, two cupcakes do not always have the same number of sprinkles on them. The number of sprinkles on each cupcake is independent and geometrically distributed with parameter  $p$ .

- a. What is the probability that both cupcakes have exactly the same number of sprinkles on them?
- b. For scientific purposes, you carefully scrape the sprinkles off both cupcakes onto a plate and count them. What is the distribution of the total number of sprinkles (obtained from both cupcakes)?